# SIH 26016

## Complete System Architecture, Workflow and Flowcharts

Evidence-Backed National Land Acquisition Control Plane

**State systems remain authoritative. The platform proves legal, financial, physical and human readiness before declaring land construction-ready.**

Designed for nationwide scale, State-specific rules, offline field work, auditable evidence, compensation reconciliation, R&R safeguards and constructible-front intelligence.

**05 September 2026**

# 1. Architecture objective

## The central design decision

This is **not a replacement land-record database** and not merely a dashboard. State land, registration, court, finance and clearance systems remain authoritative. The national platform performs four jobs:

1. **Exchange:** ingest heterogeneous State and ministry data through governed adapters.
2. **Prove:** preserve source evidence, history, signatures, contradictions and applicable rules.
3. **Coordinate:** route long-running statutory and human workflows with named responsibility.
4. **Decide:** compute verified readiness, corridor continuity, payment exceptions and family exclusion risks.

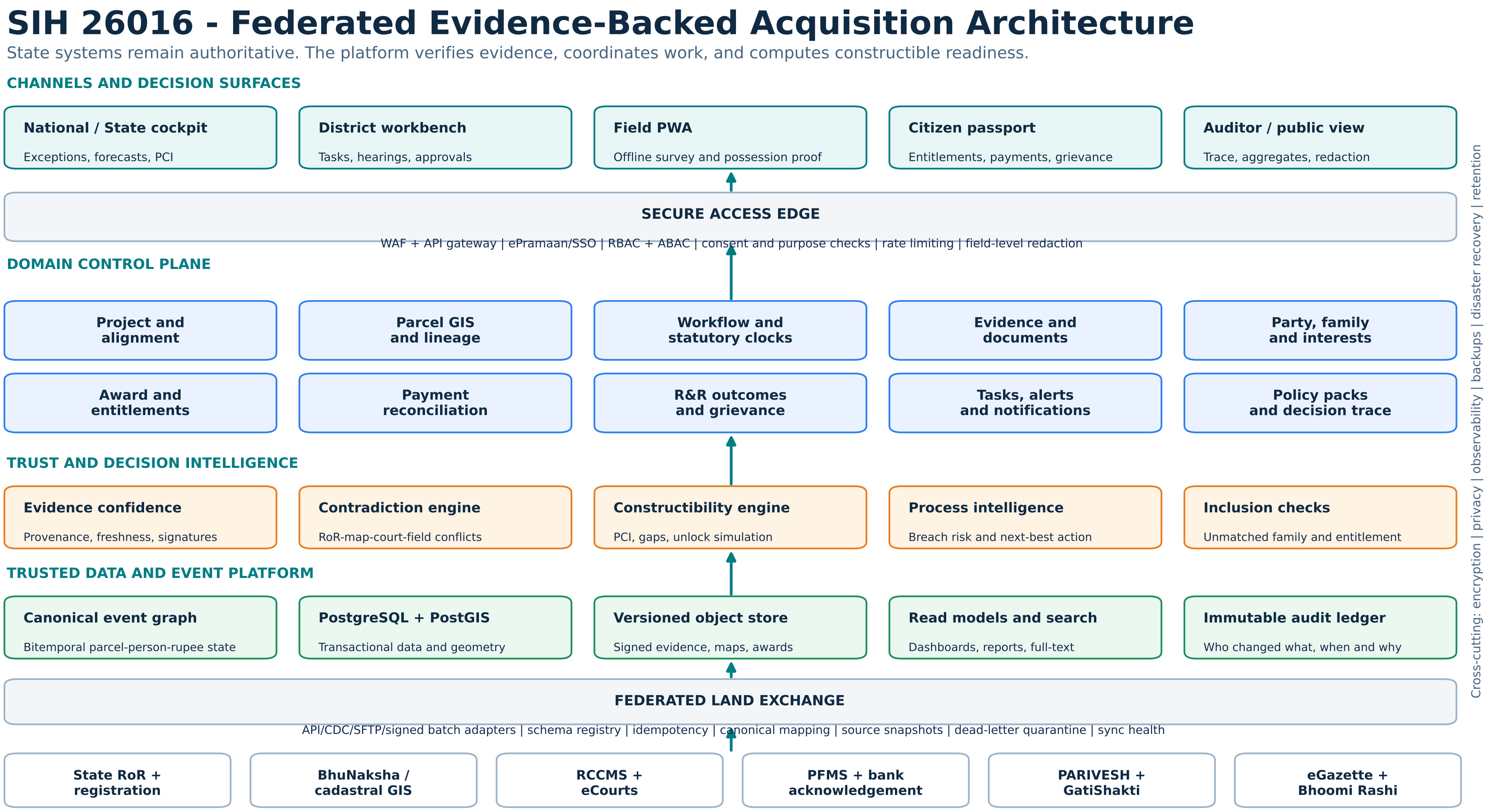
## Non-negotiable invariants

- **No status without evidence:** a dropdown cannot make a parcel acquired, paid or possessed.
- **No destructive overwrite:** corrections create new events and preserve previous facts.
- **No single project state:** parcel, claimant, payment and R&R states advance independently.
- **No silent conflict merge:** contradictory authoritative sources create an exception.
- **No AI adjudication:** AI prioritises and explains; humans decide title, compensation and eligibility.
- **No central PII exposure:** use purpose-bound access, field-level redaction and public aggregates.
- **No forced State migration:** adapters translate source models into a canonical exchange model.

## System quality targets

| Quality | Architecture response |
|---|---|
| Nationwide scalability | Stateless services, partitioning by State/project, asynchronous events, read projections |
| Long-running workflows | Durable state machine, timers, retries, human tasks, effective-dated rule packs |
| Legal auditability | Bitemporal events, signed evidence, immutable audit trail, versioned decisions |
| GIS performance | PostGIS, vector tiles, geometry simplification, cached national aggregates |
| Interoperability | Versioned APIs, schema registry, source snapshots, adapter certification |
| Field resilience | Offline-first PWA, local encrypted queue, idempotent sync, conflict handling |

# 2. Complete logical architecture




**Read bottom to top:** external government rails feed a federated exchange; canonical events and evidence power domain workflows; intelligence produces verified decisions; role-specific surfaces expose only necessary data.

# 3. Component responsibilities

## Experience and access

| Component | Primary users | Responsibility |
|---|---|---|
| National/State cockpit | Ministries, State nodal teams, policy makers | Exceptions, PCI, forecasts, comparison and escalation - no raw PII |
| District workbench | Collector, LAA, survey/revenue, legal, R&R, finance | Work queues, scrutiny, hearings, evidence correction, approvals and SLA ownership |
| Field PWA | Surveyor, possession team, R&R verifier | Offline surveys, geo/time evidence, household verification and later sync |
| Citizen passport | Affected person/family | Parcel case, entitlements, payment state, grievance, appointments and consent |
| Audit/public view | Auditor, authorised oversight, public | Full trace for auditors; redacted aggregate transparency for public users |

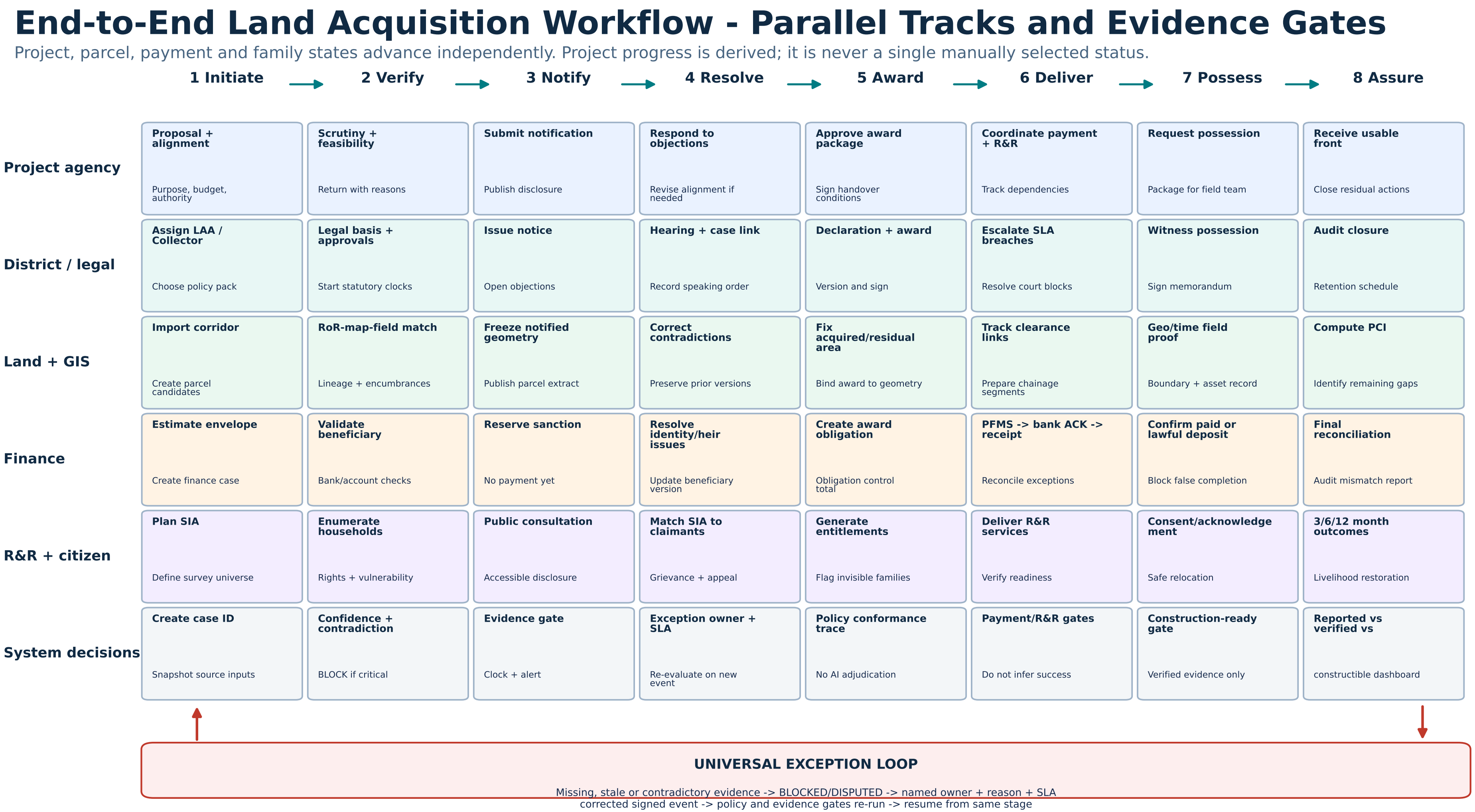
## Domain modules

| Module | Owns | Emits important events |
|---|---|---|
| Project and alignment | Project authority, corridor, chainage, funding, phases | `ProjectSubmitted`, `AlignmentVersioned` |
| Parcel GIS and lineage | Parcel versions, ULPIN/source IDs, split/merge, acquired/residual areas | `ParcelMatched`, `ParcelSplit`, `GeometryFrozen` |
| Workflow and clocks | State machine, tasks, hearings, due dates, blockers | `TaskAssigned`, `GateBlocked`, `DeadlineBreached` |
| Evidence and documents | Evidence metadata, versions, signatures, retention, access labels | `EvidenceAdded`, `EvidenceSuperseded`, `SignatureFailed` |
| Party/family/interests | Person, household, rights, relationship, vulnerability, representation | `HouseholdObserved`, `InterestVerified`, `PossibleExclusion` |
| Award and entitlement | Valuation snapshot, award, entitlement lines, legal basis | `AwardDeclared`, `EntitlementCreated` |
| Payment reconciliation | Obligation, sanction, payment instruction, bank ACK, receipt/exception | `PaymentAcknowledged`, `PaymentFailed`, `PaymentReconciled` |
| R&R and grievance | R&R plan, services, site readiness, outcomes, grievance/appeal | `RNRDelivered`, `OutcomeVerified`, `GrievanceEscalated` |
| Policy packs | Effective rules, evidence requirements, clocks and decision tables | `PolicyEvaluated`, `ConformanceFailed` |
| Notifications | Template, channel, delivery status, retry, preferred language | `NoticeQueued`, `NoticeDelivered`, `NoticeFailed` |

## Recommended implementation boundary

For SIH, implement these as a **modular monolith** with one PostgreSQL/PostGIS database and a transactional outbox. Keep clean domain modules and versioned APIs. Split high-load components - GIS tiles, documents, notifications, analytics and ML - only in the production scale path. This is credible and avoids a fragile weekend microservice demo.

# 4. End-to-end acquisition workflow




**Important:** legal, GIS, finance and R&R tracks run in parallel but converge at evidence gates. Any gate failure creates an owned exception and resumes from the same stage after corrected evidence arrives.

# 5. Workflow stage specification

## Stage 1 - Initiate

- **Input:** project purpose, requiring authority, approved/preliminary corridor, funding and applicable acquisition regime.
- **Actions:** create immutable case ID; select State/jurisdiction policy pack; import alignment; identify candidate parcels; initialise SIA and finance cases.
- **Gate:** authority, purpose, geometry and required proposal documents present.
- **Output:** submitted project version and parcel-candidate inventory.

## Stage 2 - Verify

- Cross-check RoR, registration, cadastral geometry, court/revenue cases and field observations.
- Build parcel lineage and party-interest graph; enumerate affected households and vulnerabilities.
- Generate contradiction cases. Critical conflicts block only the affected parcel/stage, not the entire platform.
- **Output:** evidence confidence profile and verified/not-verified parcel inventory.

## Stage 3 - Notify

- Freeze notified geometry and source snapshot; issue statutory notice; publish accessible citizen extract.
- Start effective statutory clock and delivery tracking; record failed service and re-service action.
- Open objection, hearing and grievance channels linked to the exact notification version.

## Stage 4 - Resolve

- Conduct hearings; link RCCMS/eCourt matters; record speaking orders and corrections.
- Resolve ownership/heir, geometry and family-enumeration contradictions without rewriting prior versions.
- Re-run policy and evidence gates after every corrective event.

## Stage 5 - Award

- Freeze acquired/residual geometry, valuation inputs, affected-interest set and entitlement schedule.
- Generate digitally signed award version and obligation control total.
- Block award finalisation if a high-risk unmatched household or unresolved geometry/title contradiction remains.

## Stage 6 - Deliver

- Send payment instruction; reconcile sanction, PFMS response, bank acknowledgement and beneficiary receipt/lawful deposit.
- Deliver R&R services and independently verify site/service readiness.
- A generated instruction is **not** payment completion; a service invoice is **not** an R&R outcome.

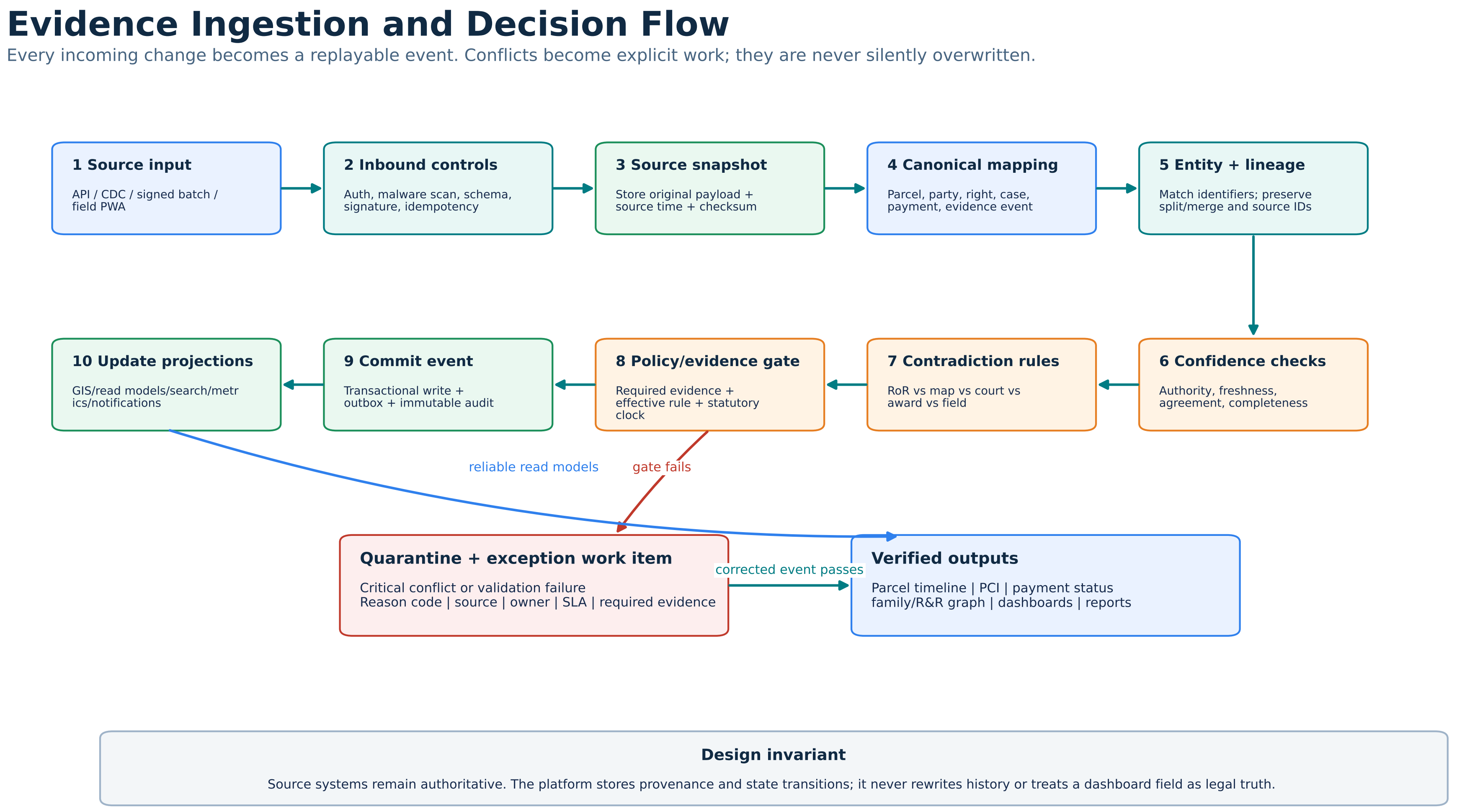
## Stage 7 - Possess

- Field team captures offline geo/time evidence, parcel boundary, assets, witnesses and signed possession memorandum.
- Confirm applicable payment/R&R/clearance dependencies; create signed proof-of-possession package.
- Mark parcel construction-ready only after all applicable gates pass.

## Stage 8 - Assure and close

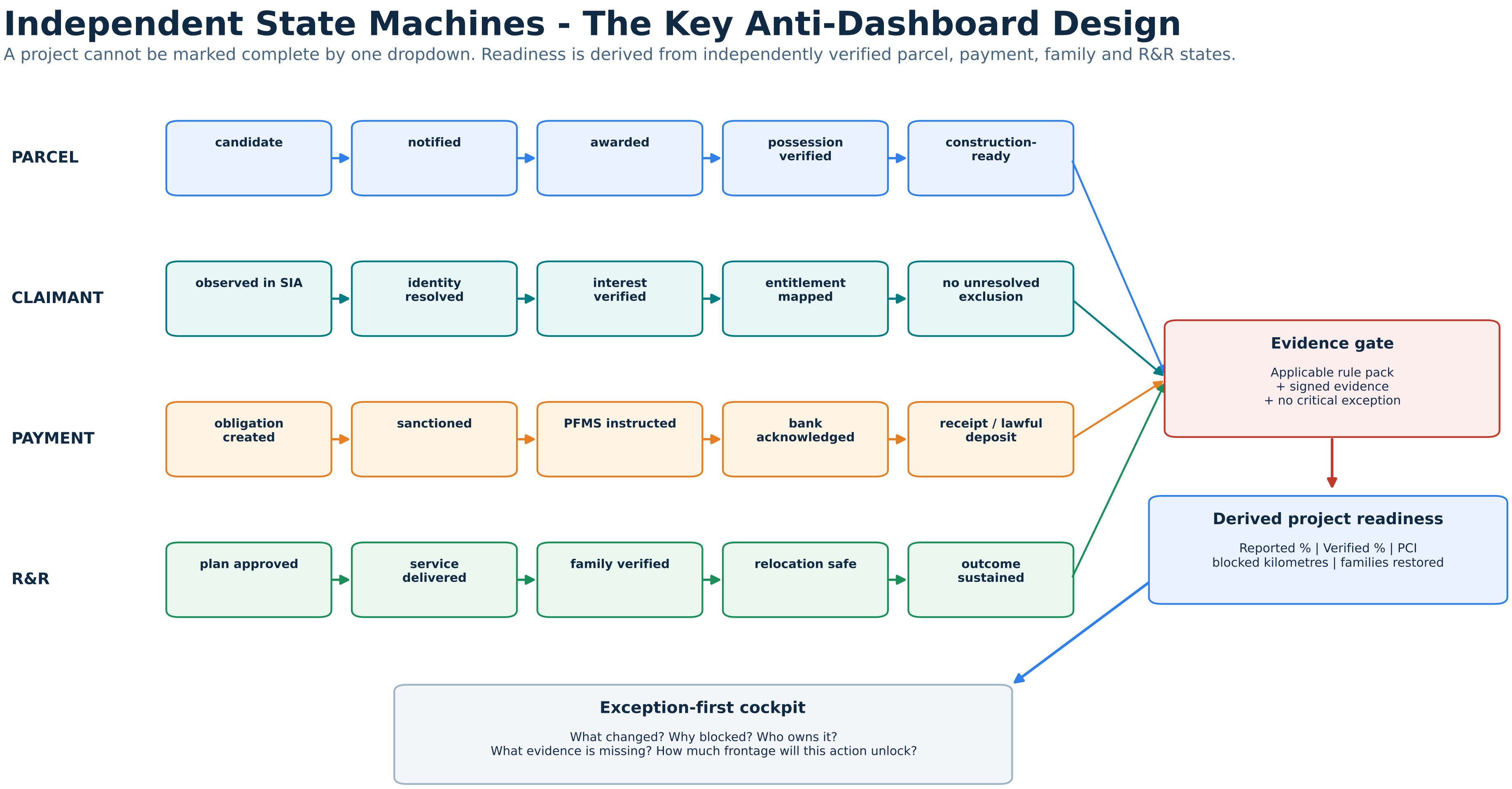
- Compute PCI, total buildable length, blocked gaps and highest-unlock action.
- Handover verified continuous fronts to the implementing agency.
- Continue 3/6/12-month family outcomes, grievances, residual parcel issues and final financial reconciliation.
- Close only after audit pack, retention classification and unresolved exceptions are accepted by authorised roles.

# 6. Evidence ingestion and decision flow




**Critical behaviour:** a failed mapping, signature, confidence rule or policy gate goes to quarantine with reason, owner, SLA and required evidence. The source payload remains intact for replay and audit.

# 7. Independent state machines and derived readiness




**Why this matters:** ”project complete” is a calculated view. A parcel may be possessed while one payment is failed or a family outcome remains unresolved; the dashboard must expose that truth instead of collapsing it.

# 8. Canonical event and data model

## Core entities

| Entity | Essential fields | Critical relationship |
|---|---|---|
| `Project` | project ID, authority, purpose, jurisdiction, policy pack, dates | has alignment versions and acquisition cases |
| `AlignmentSegment` | geometry, chainage start/end, version, clearance links | intersects parcel versions |
| `ParcelVersion` | ULPIN/source IDs, geometry, area, valid/recorded times, provenance | parent/split/merge; supports acquired/residual portions |
| `Party` / `Household` | tokenised ID, relationship, communication preference, vulnerability | holds one or more interests; appears in observations |
| `Interest` | owner/tenant/livelihood/FRA/heir type, share, evidence | connects party/household to parcel |
| `SIAObservation` | survey event, enumerator, location, finding, evidence | creates affected-family universe |
| `Case` | objection/court/revenue/grievance type, status, order, due date | blocks specific evidence gate when applicable |
| `AwardVersion` | legal basis, valuation inputs, geometry snapshot, signature | creates entitlement obligations |
| `Entitlement` | beneficiary, component, amount/service, eligibility basis | connects award to payment or R&R service |
| `PaymentEvent` | instruction, PFMS ref, bank ACK, amount, failure reason | reconciles against entitlement obligation |
| `PossessionEvent` | parcel version, geo/time proof, witness, memorandum | supports construction-ready gate |
| `RNROutcome` | service/outcome, due date, verification, follow-up | connects family to sustained outcome |
| `EvidenceArtifact` | type, hash, signer, source, access class, supersedes | supports any entity/event without becoming the fact itself |
| `WorkflowEvent` | event type, actor, source time, recorded time, causation ID | immutable source for state projections |
| `PolicyPack` | jurisdiction, legal regime, effective dates, rules, evidence matrix | evaluates events at the correct historical rule version |

## Event envelope

Every integration and internal mutation uses the same envelope:

`event_id, event_type, aggregate_type, aggregate_id, source_system, source_record_id, source_time, recorded_time, schema_version, policy_version, actor, correlation_id, causation_id, payload, evidence_refs, signature/checksum, sensitivity_label`

Rules:

- Global uniqueness plus **idempotency key** prevents duplicates during retry.
- `source_time` and `recorded_time` provide bitemporal auditability.
- Corrections reference `supersedes_event_id`; old events remain queryable.
- PII is tokenised; sensitive evidence is fetched only after role, purpose and case checks.

# 9. Core algorithms

## Possession Continuity Index (PCI)

1. Intersect the approved alignment with parcel geometries and convert results into ordered chainage intervals.
2. Evaluate `ready(interval)` from applicable parcel, payment, possession, R&R and clearance gates.
3. Merge adjacent ready intervals; calculate the longest uninterrupted ready run.
4. Rank each blocking parcel by simulated increase in continuous ready length if its exception is resolved.

$$PCI = \frac{\text{max}\left(\text{continuous evidence-ready run length}\right)}{\text{total alignment length}}$$

Display **total acquired area**, **verified area**, **total buildable length**, **longest continuous front**, **PCI**, **blocked kilometres** and **highest-unlock parcel** separately.

## Evidence confidence and contradiction

Confidence is explanatory, not a legal verdict. Show the component reasons:

- Source authority and signature validity.
- Freshness against source-specific tolerance.
- Cross-source agreement on identity, geometry, ownership/interest and amount.
- Completeness of mandatory evidence under the active policy pack.
- Field evidence quality and independent confirmation.

Hard contradictions - invalid parcel ID, geometry mismatch above tolerance, active blocking case, award/payment mismatch, duplicate reference, missing signature - create deterministic exceptions regardless of score.

## No Family Invisible rule

```
Affected-family universe = union(SIA observations, field verification, recorded interests,
claims/objections and recognised livelihood dependencies)
```

Flag every household with no resolved mapping to an award decision, entitlement/R&R decision and communicated reason. “Not eligible” is a valid reviewed outcome; **disappearing from the data is not**.

## Delay intelligence

Start with rule-based statutory/SLA risk. When sufficient real event history exists, use survival analysis/process mining to estimate time-to-next-stage. Always expose evidence, uncertainty and recommended human action. Never train a black-box model on self-reported status snapshots.

# 10. Integration architecture

## Adapter contract

Every source adapter must implement:

- `pullSnapshot(cursor)` or authenticated inbound event endpoint.
- Source schema validation and schema-version declaration.
- Stable source record ID and idempotency key.
- Signed payload/checksum and source timestamp.
- Canonical mapping with unmapped-field preservation.
- Health, last-success, lag and rejection metrics.
- Replay from cursor plus dead-letter reprocessing.

## Integration modes

| Source capability | Integration mode | Safety mechanism |
|---|---|---|
| Mature API | REST/event webhook | OAuth/mTLS, idempotency, contract tests |
| Database change feed | CDC into connector | read-only replica, checkpoint, source lag |
| Periodic export | Signed SFTP/batch | manifest, checksum, schema version, replay |
| No digital interface | Authorised manual upload | digital signature, maker-checker, mandatory provenance |
| Field offline | Encrypted sync queue | device binding, idempotent batch, conflict review |

## External systems to integrate, not rebuild

- State RoR, registration and cadastral/BhuNaksha systems; preserve State identifiers and ULPIN where available.
- RCCMS/eCourt for pending land/revenue disputes.
- PFMS and bank acknowledgements for reconciliation.
- PARIVESH and approved PM GatiShakti alignment/clearance references.
- e-Gazette/Bhoomi Rashi or ministry-specific acquisition systems.

For SIH, demonstrate **three realistic mock adapters** - land record/cadastral, court case and PFMS/bank - with the exact envelope, source-health status and contradiction handling. Do not claim live government access.

# 11. Security, privacy and governance

## Identity and authorisation

- Government SSO/ePramaan-compatible identity boundary; MFA for privileged roles.
- RBAC for organisational responsibility plus ABAC for State, district, project, case, purpose and data sensitivity.
- Maker-checker for awards, payment changes, possession and policy-pack publication.
- Short-lived tokens, mTLS service identity, secrets manager and device binding for field sync.

## Data protection

- TLS in transit; envelope encryption at rest; centrally governed key rotation.
- PII tokenisation and separate identity vault; dashboards query pseudonymous IDs.
- Field-level redaction; public views contain aggregates and generalised geometries only.
- Consent/preference is recorded where relevant, but statutory processing authority is represented separately.
- Configurable retention and legal hold; secure deletion after approved retention expiry.

## Evidence and audit

- Versioned object storage with object lock/WORM for final evidence packages.
- Hash and digital signature prove integrity; blockchain is unnecessary unless cross-agency key governance is explicitly solved.
- Immutable audit event: actor, purpose, old/new reference, reason, timestamp, device/service and correlation ID.
- Every automated suggestion records model/rule version, inputs, confidence and human disposition.

## Availability and operations

- OpenTelemetry tracing, metrics and structured logs; sync-lag and dead-letter alerts.
- Point-in-time database recovery, replicated object storage, tested backup restoration and regional DR plan.
- Graceful degradation: field collection continues offline; integrations retry without duplicate state transitions.

# 12. Deployment and technology stack

## Hackathon implementation

| Layer | Recommended choice | Why |
|---|---|---|
| Web | Next.js + TypeScript | Fast dashboard/workbench development; SSR where useful |
| Maps | OpenLayers or MapLibre + GeoServer | Standards-based parcel and vector-layer handling |
| Field | Installable PWA with encrypted IndexedDB | One codebase; offline queue and later synchronisation |
| Backend | NestJS modular monolith | Typed modules, guards, APIs and background jobs |
| Transaction/GIS | PostgreSQL + PostGIS | Strong transactions, spatial joins and lineage |
| Documents | MinIO/S3-compatible versioned object store | Signed evidence and immutable final packages |
| Jobs/events | Transactional outbox + Redis/BullMQ | Reliable MVP workflows without Kafka complexity |
| Intelligence | Python FastAPI service | PCI/geospatial computation and explainable models |
| Auth | Keycloak or equivalent OIDC provider | Real RBAC/SSO demo instead of hard-coded roles |
| Observability | OpenTelemetry + Prometheus/Grafana | Trace cross-module failures and integration lag |

## Production evolution

- Deploy containers on NIC Cloud/MeghRaj-compatible Kubernetes.
- Move durable human workflow/timers to Camunda/Temporal; publish domain events through Kafka.
- Separate GIS tile, document, notification and analytics workloads when measured load requires it.
- Partition large event/projection tables by State and project; keep national aggregates in refreshed read models.
- Add OpenSearch for document/full-text search and a dedicated analytics store only after query patterns justify it.

## Deployment topology

```
Internet/Gov network -> WAF/load balancer -> stateless app/API pods -> workflow/domain modules
-> PostgreSQL/PostGIS + object store + cache/outbox -> adapters through controlled egress/API
gateway
```

Administrative and evidence stores stay on private subnets. Public aggregate views use separate read projections, never direct queries against case/PII tables.

# 13. MVP build sequence and demo trace

## P0 - must work end to end

1. Create project and import a synthetic 12 km alignment with parcel polygons.
2. Ingest land-record, court and payment events through three mock adapters.
3. Show source snapshots, parcel lineage and one critical contradiction.
4. Run policy/evidence gates and generate owned exception tasks.
5. Compute PCI, blocked kilometres and highest-unlock parcel.
6. Detect four SIA households missing from award/R&R mapping.
7. Reconcile one successful and one failed payment.
8. Add field possession proof and watch verified readiness update through events.

## P1 - only after P0 is polished

- Offline field queue and signed proof package.
- Citizen entitlement/payment/grievance passport.
- R&R service verification and one 3-month outcome.
- Rule-based delay explanation and notification delivery tracking.

## Exact 4-minute walkthrough

| Time | Screen | Judge-visible point |
|---:|---|---|
| 0:00-0:30 | Project cockpit | Ordinary metric says 86% acquired; verified progress is 61%; PCI is 42% |
| 0:30-1:10 | Corridor gap view | Three parcels break the front; P-118 has the highest unlock value |
| 1:10-2:00 | Parcel evidence timeline | Succession conflict, court case and source provenance explain the block |
| 2:00-2:40 | Family/entitlement graph | Four livelihood-dependent households exist in SIA but not award/R&R |
| 2:40-3:20 | Payment reconciliation | Payment instruction exists, but bank acknowledgement failed |
| 3:20-4:00 | Resolve simulation | Correct evidence raises PCI from 42% to 78%; task and audit trace are generated |

## Acceptance tests

- Replaying the same adapter batch creates no duplicate event or payment.
- Correcting a parcel preserves the old geometry and source snapshot.
- A critical contradiction prevents construction-ready state.
- Failed bank acknowledgement prevents payment-complete state.
- Unmatched SIA household remains visible until a reviewed outcome exists.
- Resolving the highest-criticality parcel changes PCI exactly as predicted.
- Public role cannot access PII or unredacted evidence.
- Offline field retry produces one possession event, not duplicates.

# 14. Final architecture narrative for judges

Data enters through federated adapters from State land records, cadastral maps, courts, finance and clearance systems. We preserve the original source snapshot, map it into a canonical parcel-person-payment event graph, and run deterministic contradiction and policy gates. Legal, GIS, finance and R&R workflows remain separate, so a project can never be marked complete by one misleading status. Verified parcel events feed our Possession Continuity Index, which shows the actual uninterrupted construction-ready corridor and the parcel whose resolution unlocks the most work. At the same time, the claimant-entitlement graph ensures no affected family disappears between SIA, award, payment and rehabilitation. Decision-makers see exceptions and accountable actions; citizens see only their own entitlement journey; auditors can reconstruct what was known when.

## Final one-line architecture

**Federated sources -> source-preserving events -> contradiction and policy gates -> independent parcel/payment/family/R&R states -> verified constructibility and accountable action.**

## Authoritative research anchors

- DoLR - DILRMP, ULPIN, NGDRS, eCourt linkage, RCCMS
- LACRRIS launch and ranking design
- Bhoomi Rashi scope
- PM GatiShakti scope
- CAG Bharatmala Audit 2023
- CAG Tamil Nadu STAR 2.0 Audit
- DoLR/NCAER - Evaluation of Quality of Land Records
- RFCTLARR Act and Rules
- World Bank ESS5
- ISO 19152-1:2024 LADM